

Solving Pair of Linear Equations :-

Elimination Method $\rightarrow$

(I) $3x - 5y = 5$ and $6x - 7y = 16$

$$\begin{array}{rcl} (3x - 5y = 5) \times 2 & - & \text{Eq}^n (1) \\ 6x - 7y = 16 & - & \text{Eq}^n (2) \end{array}$$

$$\begin{array}{r} \cancel{6x} - 10y = 10 \\ - \cancel{6x} + 7y = 16 \\ \hline -3y = -6 \end{array} \quad \{ \text{Subtracting} \}$$

$$y = \frac{6}{3} \quad y = \underline{\underline{2}}$$

Putting value of y in Equation (1)

$$3x - 5(2) = 5$$

$$3x = 5 + 10$$

$$3x = 15$$

$$\underline{\underline{x = 5}}$$

$$\boxed{\begin{array}{l} x = 5 \\ y = 2 \end{array}}$$

(II) $4x - 3y = 18$ and $6x - 5y = 22$

$$\begin{array}{rcl} (4x - 3y = 18) \times 5 & - & \text{Eq}^n (1) \\ (6x - 5y = 22) \times 3 & - & \text{Eq}^n (2) \end{array}$$

$$\begin{array}{r} 20x - 15y = 90 \\ - 18x + 15y = 66 \\ \hline 2x = 24 \end{array} \quad \{ \text{Subtracting} \}$$

$$\underline{\underline{x = 12}}$$

Putting value of x in Equation (1)

$$4(12) - 3y = 18$$

$$48 - 18 = 3y$$

$$30 = 3y$$

$$\underline{\underline{y = 10}}$$

$$\boxed{\begin{array}{l} x = 12 \\ y = 10 \end{array}}$$

$$(III) \quad 2x + 5y = 12 \quad \text{and} \quad 4x - 5y = -3$$

$$2x + 5y = 12$$

Eq (1)

$$4x - 5y = -3$$

Eq (2)

$$6x = 9$$

$$x = \frac{3}{2}$$

Putting value of x in equation (1)

$$2\left(\frac{3}{2}\right) + 5y = 12$$

$$5y = 12 - 3$$

$$5y = 9$$

$$y = \frac{9}{5}$$

$$x = \frac{3}{2}, y = \frac{9}{5}$$

$$x = 1.5, y = 1.8$$

$$(IV) \quad 3x + 2y = 25 \quad \text{and} \quad 9x - 8y = 5$$

$$(3x + 2y = 25) \times 4 \quad - \quad \text{Eqn (1)}$$

$$9x - 8y = 5 \quad - \quad \text{Eq (2)}$$

$$12x + 8y = 100$$

$$9x - 8y = 5$$

(Addition)

$$21x = 105$$

$$x = \frac{105}{21}$$

$$x = 5$$

Putting value of x in equation (2)

$$9(5) - 8y = 5$$

$$45 - 5 = 8y$$

$$40 = 8y$$

$$y = 5$$

$$x = 5$$

$$y = 5$$

$$(V) \quad 15x - 8y = 106 \quad \text{and} \quad 2x + 5y = 2$$

$$(15x - 8y = 106) \times 5 \quad - \text{Eq 1}$$

$$(2x + 5y = 2) \times 8 \quad - \text{Eq 2}$$

$$75x - 40y = 530$$

$$16x + 40y = 16$$

{ Addition }

$$91x = 546$$

$$x = \frac{546}{91}$$

$$x = 6$$

Putting value of x in equation 2

$$2(6) + 5y = 2$$

$$12 + 5y = 2$$

$$5y = -10$$

$$y = -2$$

$$\boxed{\begin{matrix} x = 6 \\ y = -2 \end{matrix}}$$

$$(VI) \quad 15x + 10y = 50 \quad \text{and} \quad 30x + 20y = 100$$

$$\cancel{5}(3x + 2y) = \cancel{5} \times 10$$

$$\cancel{30}x + \cancel{20}y = \cancel{100} \quad (\text{Simplify})$$

$$3x + 2y = 10$$

$$3x + 2y = 10$$

Same equations

Either

$$x = y$$

or

Relationship between x and y
can not be established

Comparison of Roots →

$$(I) \quad 3x^2 - 44x + 160 = 0$$

$$\text{Product} = 480$$

$$\text{Sum} = 44$$

$$\left(\frac{+24}{3}, \frac{+20}{3} \right)$$

$$(II) \quad y^2 - 19y + 88 = 0$$

$$\text{Product} = 88$$

$$\text{Sum} = 19$$

$$\left(+11, +8 \right) \times 3$$

$$(33, 24)$$

⇒ Comparison :

$$\begin{array}{l} \text{Roots of } x = 24, 20 \\ \text{Roots of } y = 33, 24 \end{array} \left\{ \begin{array}{ll} 24 < 33 & 20 < 33 \\ 24 = 24 & 20 < 24 \end{array} \right\}$$

$$\boxed{\text{So, } x \leq y}$$

$$(I) \quad 4x^2 - 8x + 3 = 0$$

$$\text{Product} = 12$$

$$\text{Sum} = 8$$

$$\left(+6, +2 \right) \times 3 \rightarrow \text{coefficient of } y^2$$

$$(18, 6)$$

$$(II) \quad 3y^2 - 16y + 5 = 0$$

$$\text{Product} = 15$$

$$\text{Sum} = 16$$

$$\left(+15, +1 \right) \times 4 \rightarrow \text{coefficient of } x^2$$

$$(60, 4)$$

⇒ Comparison :

$$\begin{array}{l} \text{Roots of } x^2 = 18, 6 \\ \text{Roots of } y^2 = 60, 4 \end{array} \left\{ \begin{array}{ll} 18 < 60 & 6 < 60 \\ 18 > 4 & 6 > 4 \end{array} \right\}$$

Both case available

$$\begin{array}{l} x > y \\ x < y \end{array}$$

So, Relation between x and y can not be established.

$$(I) \quad 25x^2 - 15x + 2 = 0$$

$$\text{Product} = 50$$

$$\text{Sum} = 15$$

$$(10, 5) \times 5 \rightarrow \text{coefficient of } y^2$$

$$(50, 25)$$

$$(II) \quad 5y^2 + 19y - 4 = 0$$

$$\text{Product} = 20$$

$$\text{Difference} = 19$$

$$(-20, 1) \times 25 \rightarrow \text{coefficient of } x^2$$

$$(-500, 25)$$

Comparison of Roots $\rightarrow$

$$\text{Roots of } x = 50, 25$$

$$\text{Roots of } y = -500, 25$$

$$\left\{ \begin{array}{ll} 50 > -500 & 25 > -500 \\ 50 > 25 & 25 = 25 \end{array} \right\}$$

$$\boxed{\text{So, } x \geq y}$$

$$(I) \quad 2x^2 - 7x + 3 = 0$$

$$\text{Product} = 6$$

$$\text{Sum} = 7$$

$$(+6, +1)$$

$$(II) \quad 2y^2 - 27y + 70 = 0$$

$$\text{Product} = 140$$

$$\text{Sum} = 27$$

$$(+20, +7)$$

Both coefficient of x^2 & y^2 are equal.

Comparison of Roots $\rightarrow$

$$\text{Roots of } x = 6, 1$$

$$\text{Roots of } y = 20, 7$$

$$\left\{ \begin{array}{ll} 6 < 20 & 1 < 20 \\ 6 < 7 & 1 < 7 \end{array} \right\}$$

$$\boxed{\text{So, } x < y}$$

$$(I) 4x^2 - 37x + 40 = 0$$

$$\text{Product} = 160$$

$$\text{Sum} = 37$$

$$(32, 5) \times (5) \text{ coefficient of } y^2$$

$$(160, 25)$$

$$(II) 5y^2 - y - 4 = 0$$

$$\text{Product} = 20$$

$$\text{Difference} = 1$$

$$(-4, +5) \times (4) \text{ coefficient of } x^2$$

$$(-16, 20)$$

Comparison of Roots $\rightarrow$

$$\begin{array}{l} \text{Roots of } x = 160, 25 \\ \text{Roots of } y = -16, 20 \end{array} \left\{ \begin{array}{ll} 160 > -16 & 25 > -16 \\ 160 > 20 & 25 > 20 \end{array} \right\}$$

$$\boxed{\text{So, } x > y}$$

$$(I) 6x^2 - 7x + 2 = 0$$

$$\text{Product} = 12$$

$$\text{Sum} = 7$$

$$(+4, +3) \times (3) \text{ coefficient of } y^2$$

$$(12, 9)$$

$$(II) 3y^2 - 20y + 12 = 0$$

$$\text{Product} = 36$$

$$\text{Sum} = 20$$

$$(+18, +2) \times (6) \text{ coefficient of } x^2$$

$$(108, 12)$$

Comparison of Roots $\rightarrow$

$$\begin{array}{l} \text{Roots of } x = 12, 9 \\ \text{Roots of } y = 108, 12 \end{array} \left\{ \begin{array}{ll} 12 < 108 & 9 < 108 \\ 12 = 12 & 9 < 12 \end{array} \right\}$$

$$\boxed{\text{So, } x \leq y}$$

$$(I) x^2 - 49x + 600 = 0$$

$$\text{Product} = 600$$

$$\text{Sum} = 49$$

$$(+24, +25)$$

$$(II) y^2 - 35y + 304 = 0$$

$$\text{Product} = 304$$

$$\text{Sum} = 35$$

$$(19, 16)$$

Comparison of Roots $\rightarrow$

$$\begin{array}{l} \text{Roots of } x = 24, 25 \\ \text{Roots of } y = 19, 16 \end{array} \left\{ \begin{array}{ll} 24 > 19 & 25 > 19 \\ 24 > 16 & 25 > 16 \end{array} \right\}$$

$$\boxed{\text{So, } x > y}$$

$$(I) 8x^2 - 22x + 5 = 0$$

$$\text{Product} = 40$$

$$\text{Sum} = 22$$

$$\left(\frac{2}{8}, \frac{20}{8} \right)$$

$$(2, 20)$$

$$(II) y^2 - 22y + 105 = 0$$

$$\text{Product} = 105$$

$$\text{Sum} = 22$$

$$(7, 15) \times 8$$

$$(56, 120)$$

Comparison of Roots $\rightarrow$

$$\begin{array}{l} \text{Roots of } x = 2, 20 \\ \text{Roots of } y = 56, 120 \end{array} \left\{ \begin{array}{ll} 2 < 56 & 20 < 56 \\ 2 < 120 & 20 < 120 \end{array} \right\}$$

$$\boxed{\text{So, } x < y}$$

Sign Method { Direct Solution }

Case I →

$$ax^2 - bx + c = 0$$

$$py^2 + qy + r = 0$$

Roots

(+, +)

(-, -)

} $x > y$

$$ax^2 + bx + c = 0$$

$$py^2 - qy + r = 0$$

(-, -)

(+, +)

} $x < y$

Case II →

$$ax^2 + bx - c = 0$$

$$py^2 + qy - r = 0$$

Roots

(+, -)

(+, -)

$$ax^2 - bx - c = 0$$

$$py^2 - qy - r = 0$$

(+, -)

(+, -)

$$ax^2 + bx - c = 0$$

$$py^2 - qy - r = 0$$

(+, -)

(+, -)

$$ax^2 - bx - c = 0$$

$$py^2 + qy - r = 0$$

(+, -)

(+, -)

In all these cases,

Relationship between x and y can not be established

In cases, where roots of x are (+, -)

Also, the roots of y are (+, -)

Means there are both $x > y$ & $x < y$ roots.

So, the relation between x & y can not be determined.

Practice Questions $\rightarrow$

(I) $5x^2 + x - 6 = 0$

Roots

 $(+, -)$

(II) $2y^2 - 3y - 2 = 0$

 $(+, -)$

Because constant
term sign $(-)$

Relationship can not be established.

(I) $0.5x^2 - 8x + 31.5 = 0$ $(+, +)$

(II) $0.25y^2 - 2.5y + 6 = 0$ $(+, +)$

(Not a case of sign Method)

(I) $16x^2 - 32x + 7 = 0$ $(+, +)$

(II) $y^2 + 76y + 1344 = 0$ $(-, -)$

$\{ x > y \}$

(I) $2x^2 - 54x + 304 = 0$ $(+, +)$

(II) $y^2 + 7y - 120 = 0$ $(+, -)$

(Not a case of sign Method)

$\Rightarrow$ We can not compare $(+, +)$ roots
and $(+, -)$ roots on the basis of
sign Method.

(I) $x^2 + 9x - 90 = 0$ $(+, -)$

(II) $2y^2 - 13y - 7 = 0$ $(+, -)$

$\{ \text{Relationship between } x \text{ and } y \}$
 $\{ \text{can not be established.} \}$

$$(I) x^2 + 18x + 80 = 0$$

$$\text{Product} = 80$$

$$\text{Sum} = 18$$

→ Same sign roots

$$* (-8, -10)$$

$$(II) y^2 + 10y + 24 = 0$$

$$\text{Product} = 24$$

$$\text{Sum} = 10$$

→ Same sign roots

$$(-4, -6)$$

Comparison of roots →

$$\begin{array}{l} \text{Roots of } x = -8, -10 \\ \text{Roots of } y = -4, -6 \end{array} \left\{ \begin{array}{ll} -8 < -4 & -10 < -4 \\ -8 < -6 & -10 < -6 \end{array} \right\}$$

$$\boxed{\text{So, } x < y}$$

$$(I) 0.5x^2 - 8x + 31.5 = 0$$

$$\left(\frac{x^2}{2} - 8x + \frac{63}{2} = 0 \right) \times 2$$

$$x^2 - 16x + 63 = 0$$

$$\text{Product} = 63$$

$$\text{Sum} = 16$$

$$(+9, +7)$$

$$(II) 0.25y^2 - 2.5y + 6 = 0$$

$$\left(\frac{y^2}{4} - \frac{10}{4}y + 6 = 0 \right) \times 4$$

$$y^2 - 10y + 24 = 0$$

$$\text{Product} = 24$$

$$\text{Sum} = 10$$

$$(+6, +4)$$

Comparison of Roots →

$$\begin{array}{l} \text{Roots of } x = 9, 7 \\ \text{Roots of } y = 6, 4 \end{array} \left\{ \begin{array}{ll} 9 > 6 & 7 > 6 \\ 9 > 4 & 7 > 4 \end{array} \right\}$$

$$\boxed{\text{So, } x > y}$$

$$(I) 2x^2 - 54x + 304 = 0$$

$$2(x^2 - 27x + 152) = 0$$

$$x^2 - 27x + 152 = 0$$

$$\text{Product} = 152$$

$$\text{Sum} = 27$$

$$(+19, +8)$$

$$(II) y^2 + 7y - 120 = 0$$

$$\text{Product} = 120$$

$$\text{Difference} = 7$$

→ Roots with opposite signs

→ Smaller root positive
(because coefficient of y is positive)

$$(+8, -15)$$

Comparison of Roots →

$$\begin{array}{l} \text{Roots of } x = 19, 8 \\ \text{Roots of } y = 8, -15 \end{array} \left\{ \begin{array}{ll} 19 > 8 & 8 = 8 \\ 19 > -15 & 8 > -15 \end{array} \right\}$$

$$\text{So, } x \geq y$$

$$(I) y^2 - 34y + 288 = 0$$

$$\text{Product} = 288$$

$$\text{Sum} = 34$$

$$(16, 18) \times 12 \text{ coefficient of } x^2$$

$$(192, 216)$$

$$(II) 12x^2 + x - 6 = 0$$

$$\text{Product} = 72$$

$$\text{Difference} = 1$$

$$(-9, +8)$$

Comparison of Roots →

$$\begin{array}{l} \text{Roots of } x = -9, 8 \\ \text{Roots of } y = 192, 216 \end{array} \left\{ \begin{array}{ll} -9 < 192 & 8 < 192 \\ -9 < 216 & 8 < 216 \end{array} \right\}$$

$$\text{So, } x < y$$

$$(I) 2x^2 + 80x + 750 = 0$$

$$2(x^2 + 40x + 375) = 0$$

$$x^2 + 40x + 375 = 0$$

$$\text{Product} = 375$$

$$\text{Sum} = 40$$

$(-25, -15)$ Roots

$$(II) y^2 - 2y - 35 = 0$$

$$y^2 - 2y - 35 = 0$$

$$\text{Product} = 35$$

$$\text{Difference} = 2$$

$$(7, -5)$$

Smaller sign $(-)$

Comparison of Roots $\rightarrow$

$$\begin{array}{l} \text{Roots of } x = -25, -15 \\ \text{Roots of } y = +7, -5 \end{array} \left\{ \begin{array}{ll} -25 < 7 & -15 < 7 \\ -25 < -5 & -15 < -5 \end{array} \right\}$$

$$\boxed{\text{So, } x < y}$$

$$(I) 6x^2 - 19x + 10 = 0$$

$$\text{Product} = 60$$

$$\text{Sum} = 19$$

$$(+15, +4)$$

$$\cancel{6} \quad \cancel{6}$$

$$(15, 4)$$

$$(II) y^2 - 7y + 10 = 0$$

$$\text{Product} = 10$$

$$\text{Sum} = 7$$

$$(+5, +2) \times 6$$

$$(30, 12)$$

Comparison of Roots $\rightarrow$

$$\begin{array}{l} \text{Roots of } x = 15, 4 \\ \text{Roots of } y = 30, 12 \end{array} \left\{ \begin{array}{ll} 15 < 30 & 4 < 30 \\ 15 > 12 & 4 < 12 \end{array} \right\}$$

So, No Relation

can be established.

$$(I) \quad x^2 - 44x + 475 = 0$$

Roots (+, +)

$$(II) \quad 25x^2 + 75y + 54 = 0$$

Roots (-, -)

$(x > y)$ { Sign Method }

$$(I) \quad 5x^2 + 128x + 75 = 0$$

Roots (-, -)

$$(II) \quad 8y^2 - 293y + 180 = 0$$

Roots (+, +)

$(x < y)$ { Sign Method }

$$(I) \quad x^2 - 45x + 486 = 0$$

Product = 486

Sum = 45

→ same sign roots
(+18, +27)

$$(II) \quad y^2 + 3y - 10 = 0$$

Product = 10

Difference = 3

→ smaller root positive
(+2, -5)

Comparison of Roots →

$$\begin{array}{l} \text{Roots of } x = 18, 27 \\ \text{Roots of } y = 2, -5 \end{array} \left\{ \begin{array}{ll} 18 > 2 & 27 > 2 \\ 18 > -5 & 27 > -5 \end{array} \right\}$$

$$\boxed{\therefore, x > y}$$

$$(I) x^2 + 9x + 8 = 0$$

$$\text{Product} = 8$$

$$\text{Sum} = 9$$

→ Same sign roots

$$(-1, -8)$$

$$(II) y^2 - 3y - 4 = 0$$

$$\text{Product} = -4$$

$$\text{Difference} = 3$$

→ smaller root (-)

$$(-1, +4)$$

Comparison of Roots →

$$\begin{array}{l} \text{Roots of } x = -1, -8 \\ \text{Roots of } y = -1, 4 \end{array} \left\{ \begin{array}{ll} -1 = -1 & -8 < -1 \\ -1 < 4 & -8 < 4 \end{array} \right\}$$

$$\boxed{\text{So, } x \leq y}$$

$$(I) 9x^2 - 30x + 25 = 0$$

$$\text{Product} = 25$$

$$\text{Sum} = 30$$

→ same sign roots

$$(+5, +5)$$

$$(II) y^2 - 8y + 15 = 0$$

$$\text{Product} = 15$$

$$\text{Sum} = 8$$

→ same sign roots

$$(+5, +3) \times (9) \text{ coefficient of } x^2$$

$$(45, 27)$$

Comparison of Roots →

$$\begin{array}{l} \text{Roots of } x = 5, 5 \\ \text{Roots of } y = 45, 27 \end{array} \left\{ \begin{array}{ll} 5 < 45 & 5 < 45 \\ 5 < 27 & 5 < 27 \end{array} \right\}$$

$$\boxed{\text{So, } x < y}$$

$$(I) x^2 + 2x - 323 = 0$$

$$\text{Product} = 323$$

$$\text{Difference} = 2$$

→ smaller sign (+)

$$(+17, -19)$$

$$(II) y^2 - 41y + 420 = 0$$

$$\text{Product} = 420$$

$$\text{Sum} = 41$$

→ same sign roots

$$(20, 21)$$

Comparison of Roots →

$$\left. \begin{array}{l} \text{Roots of } x = 17, -19 \\ \text{Roots of } y = 20, 21 \end{array} \right\} \begin{array}{ll} 17 < 20 & -19 < 20 \\ 17 < 21 & -19 < 21 \end{array}$$

$$\boxed{\text{So, } x < y}$$

$$(I) 3x^2 - 37x + 112 = 0 \quad (II) 3y^2 + 43y - 30 = 0$$

$$\text{Product} = 336$$

$$\text{Sum} = 37$$

→ both roots (+)

$$(+21, +16)$$

$$\text{Product} = 90$$

$$\text{Difference} = 43$$

→ smaller root (+)

$$(-45, +2)$$

→ coefficient of both x^2 & y^2 is 3.

Comparison of roots →

$$\left. \begin{array}{l} \text{Roots of } x = 21, 16 \\ \text{Roots of } y = -45, 2 \end{array} \right\} \begin{array}{ll} 21 > -45 & 16 > -45 \\ 21 > 2 & 16 > 2 \end{array}$$

$$\boxed{\text{So, } x > y}$$

$$(I) x^2 + 6x - 112 = 0$$

$$\text{Product} = 112$$

$$\text{Difference} = 6$$

→ smaller root (+)

$$(-14, +8)$$

$$(II) y^2 - 20y + 75 = 0$$

$$\text{Product} = 75$$

$$\text{Sum} = 20$$

→ Both roots (+)

$$(+5, +15)$$

Comparison of Roots →

$$\begin{array}{l} \text{Roots of } x = -14, 8 \\ \text{Roots of } y = 5, 15 \end{array} \left\{ \begin{array}{ll} -14 < 5 & 8 > 5 \\ -14 < 15 & 8 < 15 \end{array} \right\}$$

So, $x > y$ and $x < y$

Relation can not be established

$$(I) x^2 - 5x - 104 = 0$$

$$\text{Product} = 104$$

$$\text{Difference} = 5$$

→ smaller root (-)

$$(+13, -8)$$

$$(II) y^2 + 21y + 104 = 0$$

$$\text{Product} = 104$$

$$\text{Sum} = 21$$

→ Both roots (-)

$$(-13, -8)$$

Comparison of Roots →

$$\begin{array}{l} \text{Roots of } x = 13, -8 \\ \text{Roots of } y = -13, -8 \end{array} \left\{ \begin{array}{ll} 13 > -13 & -8 > -13 \\ 13 > -8 & -8 = -8 \end{array} \right\}$$

$$\boxed{\text{So, } x \geq y}$$

$$(1) 3x^2 + 13x - 10 = 0$$

$$\text{Product} = 30$$

$$\text{Difference} = 13$$

→ smaller sign (+)

$$(-15, +2) \times \textcircled{18} \text{ coefficient of } y^2$$

$$(-270, 36)$$

$$(2) 18y^2 - 27y + 10 = 0$$

$$\text{Product} = 180$$

$$\text{Sum} = 27$$

→ both roots (+)

$$(+15, +12) \times \textcircled{3} \text{ coefficient of } x^2$$

$$(45, 36)$$

Comparison of Roots →

$$\begin{array}{l} \text{Roots of } x^2 = -270, 36 \\ \text{Roots of } y^2 = 45, 36 \end{array} \left\{ \begin{array}{ll} -270 < 45 & 36 < 45 \\ -270 < 36 & 36 = 36 \end{array} \right.$$

$$\boxed{\text{So, } x \leq y}$$

$$(1) x^2 - 13x - 30 = 0$$

$$\text{Product} = 30$$

$$\text{Difference} = 13$$

→ smaller sign (-)

$$(+15, -2)$$

$$(11) 0.25y^2 - 3.25y + 7.5 = 0$$

$$\left(\frac{1}{4}y^2 - \frac{13}{4}y + \frac{30}{4} = 0 \right) \times 4$$

$$y^2 - 13y + 30 = 0$$

$$\text{Product} = 30$$

$$\text{Sum} = 13$$

$$(+10, +3)$$

Comparison →

$$\begin{array}{l} \text{Roots of } x^2 = 15, -2 \\ \text{Roots of } y^2 = 10, 3 \end{array} \left\{ \begin{array}{ll} 15 > 10 & -2 < 10 \\ 15 > 3 & -2 < 3 \end{array} \right.$$

So, Relation can't be established.